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Predicting shipment delays with gradient boosting: an explainable approach

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Abstract

We predict shipment delays and explain the predictions so that planners can act on them. The study combines predictive modelling with explainable AI for a logistics operator. We predict shipment delays and explain the predictions so that planners can act on them. The study combines predictive modelling with explainable AI for a logistics operator.

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Introduction

Late deliveries are costly for carriers and customers alike. Prior work relies on linear models and rarely explains its predictions. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models and rarely explains its predictions. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models and rarely explains its predictions. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models and rarely explains its predictions. Late deliveries are costly for carriers and customers alike. Prior work relies on linear models and rarely explains its predictions.

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Methods

Data

We used 48,210 consignment records from 2023 and 2024 covering 37 depots. Records with missing delivery scans were removed, leaving 44,902. We used 48,210 consignment records from 2023 and 2024 covering 37 depots. Records with missing delivery scans were removed, leaving 44,902.

Model

We trained an XGBoost classifier and compared it with logistic regression and a random forest. Hyperparameters were tuned with five-fold cross validation and class imbalance was handled with

SMOTE oversampling. We trained an XGBoost classifier and compared it with logistic regression and a random forest. Hyperparameters were tuned with five-fold cross validation and class imbalance was handled with SMOTE oversampling.

Explainability

SHAP values were computed for every prediction to rank the features that drive delays, and summarised with beeswarm plots. SHAP values were computed for every prediction to rank the features that drive delays, and summarised with beeswarm plots.

Results

XGBoost reached an F1 score of 0.81 on the held-out quarter, against 0.69 for logistic regression. Carrier history and route distance were the strongest predictors. XGBoost reached an F1 score of 0.81 on the held-out quarter, against 0.69 for logistic regression. Carrier history and route distance were the strongest predictors. XGBoost reached an F1 score of 0.81 on the held-out quarter, against 0.69 for logistic regression. Carrier history and route distance were the strongest predictors.

Discussion

The model is accurate enough to prioritise interventions. Explanations helped planners trust it. The model is accurate enough to prioritise interventions. Explanations helped planners trust it. The model is accurate enough to prioritise interventions. Explanations helped planners trust it. The model is accurate enough to prioritise interventions. Explanations helped planners trust it.

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Limitations

Training and testing used data from one operator, so the estimates may be optimistic. Independent temporal validation is recommended. Training and testing used data from one operator, so the estimates may be optimistic. Independent temporal validation is recommended.

References

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